01 · Limpieza y filtrado del dataset (@AppleSupport) con PySpark

Objetivo: partir del CSV completo de *Customer Support on Twitter* (~3 millones de tweets de varias marcas) y producir un conjunto de datos que contenga **solo** la conversación de `@AppleSupport`: lo que los clientes le escriben y lo que Apple responde.

Por qué Spark y Parquet: Spark procesa el CSV de manera distribuida utilizando un esquema explícito (`StructType`) y transformaciones nativas (`pyspark.sql.functions as F`). Al finalizar, el producto de datos filtrado se lleva al Driver con `toPandas()` y se almacena en formato **Parquet** con PyArrow, siguiendo el patrón de los laboratorios para evitar dependencias de `winutils.exe` en Windows.

Qué produce: `../data/processed/apple_support_filtrado.parquet` y una tabla de conteos paso a paso para la sección de *Resultados* de la tesis.

1. Preparación del entorno e inicialización de Spark

Configuramos las variables de entorno de Python para PySpark (`PYSPARK_PYTHON` y `PYSPARK_DRIVER_PYTHON`) y creamos la `SparkSession` en modo `local[*]` compatible con **Spark 3.5.9**.

In [15]:
```
# ============================================================
# 1. PREPARACIÓN DEL ENTORNO Y SPARK SESSION
# ============================================================
import os
import sys
from pathlib import Path
import pandas as pd
import matplotlib.pyplot as plt

from pyspark.sql import SparkSession
from pyspark.sql import functions as F
from pyspark.sql.types import StructType, StructField, StringType
from pyspark import StorageLevel

# Configuración del entorno Python para PySpark (estilo Laboratorios 6 y 7)
python_executable = sys.executable
os.environ["PYSPARK_PYTHON"] = python_executable
os.environ["PYSPARK_DRIVER_PYTHON"] = python_executable
os.environ["SPARK_LOCAL_IP"] = "127.0.0.1"

# Rutas de entrada y salida
RUTA_RAW = Path("../data/raw/twcs.csv")
RUTA_OUT = Path("../data/processed/apple_support_filtrado.parquet")
RUTA_OUT.parent.mkdir(parents=True, exist_ok=True)

BRAND = "AppleSupport"
conteo = {}

# Inicialización de SparkSession (Spark 3.5.9)
spark = (
    SparkSession.builder
    .appName("Tesis_01_Limpieza_AppleSupport")
    .master("local[*]")
    .config("spark.driver.host", "127.0.0.1")
    .config("spark.driver.bindAddress", "127.0.0.1")
    .config("spark.sql.shuffle.partitions", "8")
    .config("spark.sql.legacy.timeParserPolicy", "LEGACY")
    .getOrCreate()
)

spark.sparkContext.setLogLevel("WARN")
```

```
print("Spark:", spark.version)
print("Master:", spark.sparkContext.master)
print("Spark UI:", spark.sparkContext.uiWebUrl)
```

```
Spark: 3.5.9
Master: local[*]
Spark UI: http://127.0.0.1:4042
```

2. Ingesta con esquema explícito (`StructType`)

Definimos el esquema explícito para que los identificadores de 18-19 dígitos se lean como `StringType` sin perder precisión, activando `multiLine` y `escape` para manejar saltos de línea y comillas dentro de los tweets.

In [16]:
```
# ============================================================
# 2. INGESTA DEL CSV CON STRUCTTYPE
# ============================================================
schema_twcs = StructType([
    StructField("tweet_id", StringType(), True),
    StructField("author_id", StringType(), True),
    StructField("inbound", StringType(), True),
    StructField("created_at", StringType(), True),
    StructField("text", StringType(), True),
    StructField("response_tweet_id", StringType(), True),
    StructField("in_response_to_tweet_id", StringType(), True)
])

df_crudo = (
    spark.read
    .schema(schema_twcs)
    .option("header", True)
    .option("multiLine", True)
    .option("escape", '"')
    .csv(str(RUTA_RAW))
)

df_crudo.printSchema()
conteo["total_original"] = df_crudo.count()
print("Total de registros originales:", f"{conteo['total_original']:,}")
display(df_crudo.limit(5).toPandas())
```

```
root
 |-- tweet_id: string (nullable = true)
 |-- author_id: string (nullable = true)
 |-- inbound: string (nullable = true)
 |-- created_at: string (nullable = true)
 |-- text: string (nullable = true)
 |-- response_tweet_id: string (nullable = true)
 |-- in_response_to_tweet_id: string (nullable = true)

Total de registros originales: 2,811,774
```

	tweet_id	author_id	inbound	created_at	text	response_tweet_id	in_response_to_t
0	1	sprintcare	False	Tue Oct 31 22:10:47 +0000 2017	@115712 I understand. I would like to assist y...	2	
1	2	115712	True	Tue Oct 31 22:11:45 +0000 2017	@sprintcare and how do you propose we do that	NaN	
2	3	115712	True	Tue Oct 31 22:08:27 +0000 2017	@sprintcare I have sent several private messag...	1	
3	4	sprintcare	False	Tue Oct 31 21:54:49 +0000 2017	@115712 Please send us a Private Message so th...	3	
4	5	115712	True	Tue Oct 31 21:49:35 +0000 2017	@sprintcare I did.	4	

3. Selección y filtrado de `@AppleSupport`

Construimos dos subconjuntos y los unimos con `unionByName`:

1. **Respuestas de Apple:** `author_id == "AppleSupport"`.
2. **Tweets de clientes dirigidos a Apple:** tweets entrantes (`inbound == "true"`) que mencionan a `@AppleSupport` o a los que Apple respondió (identificados mediante un `left_semi` join).

In [17]:
```
# ============================================================
# 3. FILTRO DE LA CONVERSACIÓN DE @AppleSupport
# ============================================================
apple = (
    df_crudo
    .filter(F.col("author_id") == BRAND)
    .withColumn("rol", F.lit("apple"))
)

respondidos = (
    apple
    .select(F.col("in_response_to_tweet_id").alias("tweet_id"))
    .filter(F.col("tweet_id").isNotNull())
    .distinct()
)

clientes = df_crudo.filter(F.lower(F.col("inbound")) == "true")
menciona = clientes.filter(F.col("text").rlike("(?i)@applesupport"))
contestados = clientes.join(respondidos, on="tweet_id", how="left_semi")

clientes_apple = (
    menciona.unionByName(contestados)
    .withColumn("rol", F.lit("cliente"))
)

filtrado = apple.unionByName(clientes_apple)
conteo["tras_filtro_apple"] = filtrado.count()
print("Registros tras filtro AppleSupport:", f"{conteo['tras_filtro_apple']:,}")
```

```
Registros tras filtro AppleSupport: 311,379
```

4. Depuración y construcción temporal

Eliminamos duplicados por `tweet_id`, descartamos textos vacíos o nulos y convertimos `created_at` a timestamp probando los formatos del dataset.

In [18]:
```
# ============================================================
# 4. DEPURACIÓN Y DIMENSIÓN TEMPORAL
# ============================================================
d = filtrado.dropDuplicates(["tweet_id"])
conteo["tras_quitar_duplicados"] = d.count()

d = d.filter(F.col("text").isNotNull() & (F.trim(F.col("text")) != ""))
conteo["tras_quitar_vacios"] = d.count()

col_fecha = F.col("created_at")
d = d.withColumn(
    "fecha",
    F.coalesce(
        F.to_timestamp(col_fecha, "EEE MMM dd HH:mm:ss Z yyyy"),
        F.to_timestamp(col_fecha, "dd/MM/yyyy HH:mm:ss"),
        F.to_timestamp(col_fecha)
    )
)

conteo["fechas_invalidas"] = d.filter(F.col("fecha").isNull()).count()
d = d.filter(F.col("fecha").isNotNull())
print("Registros válidos con fecha:", f"{d.count():,}")
```

```
Registros válidos con fecha: 232,973
```

5. Normalización del texto y persistencia

Creamos `texto_limpio` con funciones nativas de Spark (`F.regexp_replace`, `F.trim`, `F.size`, `F.split`) y persistimos el DataFrame con `StorageLevel.MEMORY_AND_DISK` para reutilizarlo en las validaciones y en la exportación final.

In [19]:
```
# ============================================================
# 5. NORMALIZACIÓN DEL TEXTO Y PERSISTENCIA
# ============================================================
texto_norm = F.regexp_replace(F.col("text"), "[’‘`]", "'")
texto_norm = F.regexp_replace(texto_norm, r"https?://\S+", "<URL>")
texto_norm = F.regexp_replace(texto_norm, r"@(?!(?i:applesupport)?)\w+", "@USER")
texto_norm = F.trim(F.regexp_replace(texto_norm, r"\s+", " "))

dataset_preparado = (
    d
    .withColumn("texto_limpio", texto_norm)
    .withColumn("n_palabras", F.size(F.split(F.col("texto_limpio"), " ")))
    .select(
        "tweet_id",
        "rol",
        "author_id",
        "fecha",
        "text",
        "texto_limpio",
        "n_palabras",
        "in_response_to_tweet_id"
    )
)

dataset_preparado.persist(StorageLevel.MEMORY_AND_DISK)
print("Registros preparados en caché:", f"{dataset_preparado.count():,}")
display(dataset_preparado.limit(5).toPandas())
```

```
Registros preparados en caché: 232,973
```

	tweet_id	rol	author_id	fecha	text	texto_lim
0	1000003	cliente	497667	2017-11-05 05:33:26	@AppleSupport https://t.co/OnHabGwPo7	@U <U
1	1000023	apple	AppleSupport	2017-11-03 16:45:53	@356755 We'd be happy to look this over with y...	@USER W be happ look over y
2	1000077	apple	AppleSupport	2017-11-03 16:57:27	@356765 We're here to help. What happens when ...	@U We're h to h W happ when
3	1000092	cliente	356770	2017-11-03 12:52:55	@AppleSupport your c/s sucks, I was just disco...	@USER y c/s suc was disconne
4	1000186	cliente	356792	2017-11-03 12:43:38	@115858 @AppleSupport y'all need to fix this ...	@U @USER need t this sho

In [20]:
```
# Ver datos de fecha que contiene 2016
# Ver datos con fecha en 2016
dataset_preparado.filter(F.year("fecha") == 2016).show(truncate=False)
```

```
+--------+-------+------------+------------------+--------------------------------
--------------------------------------------------------------------------------
--------------------------------------------------+---------------------------------
--------------------------------------------------------------------------------
-----------------+----------+----------------------+
|tweet_id|rol    |author_id   |fecha              |text
|texto_limpio
|n_palabras|in_response_to_tweet_id|
+--------+-------+------------+------------------+--------------------------------
--------------------------------------------------------------------------------
--------------------------------------------------+---------------------------------
--------------------------------------------------------------------------------
-----------------+----------+----------------------+
|1254076 |cliente|414170      |2016-09-30 21:36:50|@AppleSupport oh wow!! I didn't k
now the phone could tell how hard I was pushing down. That's amazing, thanks!
|@USER oh wow!! I didn't know the phone could tell how hard I was pushing down. Tha
t's amazing, thanks!                                  |19        |1254073
|
|1254077 |cliente|414170      |2016-09-30 21:04:23|@AppleSupport iPhone 7, 10.0.2
|@USER iPhone 7, 10.0.2
|4         |1254081               |
|2840102 |cliente|790357      |2016-04-04 18:04:17|@AppleSupport i was just wonderin
g how you get rid of the padlock icon at the top right of the screen? https://t.co/8
YQiB4R4sl                                      |@USER i was just wondering how you g
et rid of the padlock icon at the top right of the screen? <URL>
|21        |2840101               |
|2512288 |cliente|715999      |2016-11-08 16:52:54|Anyone else getting the voice con
trol bug on ios10? It's driving me insane! #ios10bugs #voicecontrolbug @applesupport
|Anyone else getting the voice control bug on ios10? It's driving me insane! #ios10b
ugs #voicecontrolbug @USER                            |16        |NULL
|
|1254073 |apple  |AppleSupport|2016-09-30 21:34:28|@52822 In iOS 10, touch and hold
the app lightly until it wiggles. If the app doesn't wiggle, make sure that you're n
ot pressing too hard.                          |@USER In iOS 10, touch and hold the
app lightly until it wiggles. If the app doesn't wiggle, make sure that you're not p
ressing too hard. |26        |1254077               |
|1254082 |cliente|414170      |2016-09-30 20:54:03|@AppleSupport how do I delete app
s if this is what shows when I press and hold an app https://t.co/R47Tz6nsA7
|@USER how do I delete apps if this is what shows when I press and hold an app <URL>
|19        |NULL                  |
|2833060 |apple  |AppleSupport|2016-03-03 07:00:08|Welcome to... https://t.co/EZA8eR
ycDs
|Welcome to... <URL>
|3         |NULL                  |
|2058730 |apple  |AppleSupport|2016-11-06 08:59:31|@608890 We're happy to assist. Th
is article can help you to unlock your account: https://t.co/HH37urdbz9 DM us with a
ny questions. https://t.co/GDrqU22YpT          |@USER We're happy to assist. This ar
ticle can help you to unlock your account: <URL> DM us with any questions. <URL>
|21        |2058732               |
|2507390 |cliente|714875      |2016-09-06 18:44:16|@AppleSupport ugh.. Help! My MacB
ook Air thinks I have a Bluetooth Keyboard and trackpad are frozen #applehelp http
s://t.co/9RzVI3XaFs                              |@USER ugh.. Help! My MacBook Air t
hinks I have a Bluetooth Keyboard and trackpad are frozen #applehelp <URL>
|18        |NULL                  |
|2921550 |apple  |AppleSupport|2016-10-06 08:44:00|@808566 We'll happy to work with
you and answer your questions. Please DM us where we have more room. https://t.co/GD
rqU22YpT                                       |@USER We'll happy to work with you a
nd answer your questions. Please DM us where we have more room. <URL>
|20        |2921551               |
|363663  |cliente|202183      |2016-03-03 19:19:41|@AppleSupport Howdy. Today I star
ted getting a "No Sim Card Installed" message on my iPhone 6. I'm running 9.2.1\n???
|@USER Howdy. Today I started getting a "No Sim Card Installed" message on my iPhone
6. I'm running 9.2.1 ???                              |20        |NULL
|
|1254081 |apple  |AppleSupport|2016-09-30 21:03:26|@52822 We can take a look at it.
Which device and what version of iOS are you currently using?
|@USER We can take a look at it. Which device and what version of iOS are you curren
tly using?                                            |19        |1254082
|
|1282405 |cliente|420011      |2016-12-25 13:07:10|@AppleSupport-last night my watch
went completely dead for the first time. I put it on the charger overnight and I sti
ll only see this.#help https://t.co/4HewSFc7bO|@USER-last night my watch went comple
tely dead for the first time. I put it on the charger overnight and I still only see
```

```
this.#help <URL>|25        |NULL                   |
|148093  |apple  |AppleSupport|2016-04-09 19:27:30|@149685 This will provide more in
formation on the picture you are seeing: https://t.co/WIvZlsk5oQ DM us if you need a
dditional help.                                  |@USER This will provide more informa
tion on the picture you are seeing: <URL> DM us if you need additional help.
|20        |148094                 |
|2058732 |cliente|608890      |2016-11-06 01:35:52|@AppleSupport My icloud account h
as been locked for security reasons. Even my Apple ID is locked. I can't receive my
confirmation code.                               |@USER My icloud account has been loc
ked for security reasons. Even my Apple ID is locked. I can't receive my confirmatio
n code.           |22        |NULL                   |
|2840101 |apple  |AppleSupport|2016-04-04 18:03:31|@790357 We're happy to help with
your iPhone. DM us the model, iOS\n Version and the issue and we can go from there.
|@USER We're happy to help with your iPhone. DM us the model, iOS Version and the is
sue and we can go from there.                          |23        |2840103
|
|2840103 |cliente|790357      |2016-04-04 17:46:28|@AppleSupport Awrite av got a pro
blem with my iPhone
|@USER Awrite av got a problem with my iPhone
|9         |NULL                   |
|1254078 |apple  |AppleSupport|2016-09-30 21:47:26|@52822 We're so glad to have help
ed you.
|@USER We're so glad to have helped you.
|8         |1254076                |
|2921551 |cliente|808566      |2016-10-06 04:25:20|@AppleSupport Is family shared iC
loud storage a functionality I can't find or is this something that doesn't exist (y
et)?                                             |@USER Is family shared iCloud storag
e a functionality I can't find or is this something that doesn't exist (yet)?
|19        |NULL                   |
|363661  |apple  |AppleSupport|2016-03-04 14:44:00|@202183 We would love to help fix
this. Follow these steps and DM us if you need further assistance: https://t.co/UTP2
qBzCA0                                         |@USER We would love to help fix this.
Follow these steps and DM us if you need further assistance: <URL>
|20        |363663                 |
+--------+-------+------------+-------------------+-------------------------------
--------------------------------------------------------------------------------
-----------------------------------------------+-----------------------------------
--------------------------------------------------------------------------------
-----------------+----------+----------------------+
```

6. Resumen para la sección de Resultados

Documentamos cuántos tweets había al inicio, cuántos quedaron tras cada etapa y el desglose entre clientes y respuestas de Apple.

In [21]:
```
# ============================================================
# 6. RESUMEN DE CONTEOS Y VALIDACIÓN
# ============================================================
resumen_roles = (
    dataset_preparado
    .groupBy("rol")
    .count()
    .toPandas()
    .set_index("rol")["count"]
    .to_dict()
)

rango_fechas = (
    dataset_preparado
    .agg(
        F.min("fecha").alias("inicio"),
        F.max("fecha").alias("fin")
    )
    .collect()[0]
)

conteo["tweets_clientes"] = int(resumen_roles.get("cliente", 0))
conteo["respuestas_apple"] = int(resumen_roles.get("apple", 0))
conteo["fecha_inicio"] = str(rango_fechas["inicio"])
conteo["fecha_fin"] = str(rango_fechas["fin"])
```

```
df_resumen = pd.Series(conteo, name="valor").to_frame()
display(df_resumen)
```

	valor
total_original	2811774
tras_filtro_apple	311379
tras_quitar_duplicados	232973
tras_quitar_vacios	232973
fechas_invalidas	0
tweets_clientes	126113
respuestas_apple	106860
fecha_inicio	2016-03-03 07:00:08
fecha_fin	2017-12-03 17:12:28

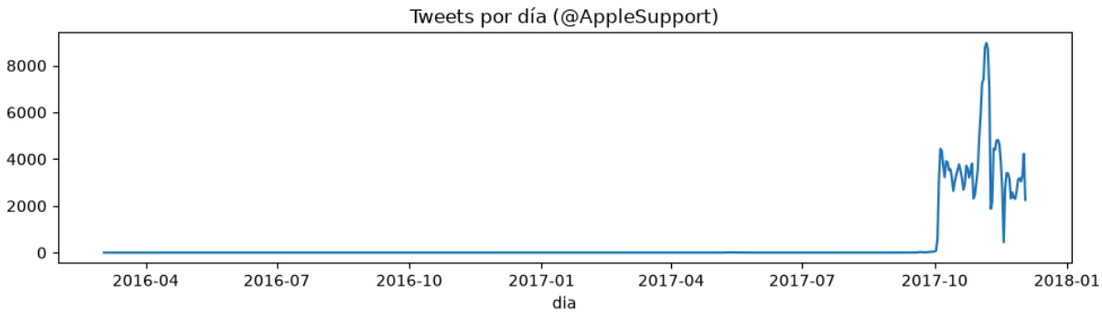
7. Exploración rápida

Estadísticos de longitud de palabras y serie diaria de volumen de tweets.

In [22]:
```
# ============================================================
# 7. EXPLORACIÓN DE LONGITUD Y VOLUMEN DIARIO
# ============================================================
dataset_preparado.select("n_palabras").describe().show()

por_dia = (
    dataset_preparado
    .groupBy(F.to_date("fecha").alias("dia"))
    .count()
    .orderBy("dia")
    .toPandas()
)
por_dia.set_index("dia")["count"].plot(figsize=(10, 3), title="Tweets por día (@App
plt.tight_layout()
plt.show()
```

```
+-------+-----------------+
|summary|       n_palabras|
+-------+-----------------+
|  count|           232973|
|   mean|20.37012014267748|
| stddev|9.259212279520792|
|    min|                1|
|    max|               94|
+-------+-----------------+
```



8. Salida en Parquet, verificación y cierre de sesión

Convertimos el producto de datos filtrado a pandas para guardarlo en formato Parquet con compresión `snappy`, verificamos la lectura y liberamos los recursos de Spark.

In [23]:
```
# ============================================================
# 8. GUARDADO EN PARQUET Y CIERRE DE SPARK
# ============================================================
dataset_preparado_pd = dataset_preparado.toPandas()
dataset_preparado_pd.to_parquet(
    RUTA_OUT,
    index=False,
    engine="pyarrow",
    compression="snappy",
    coerce_timestamps="us",
    allow_truncated_timestamps=True
)

df_verificacion = pd.read_parquet(RUTA_OUT)
print("Ruta de salida:", RUTA_OUT.resolve())
print("Registros enviados:", len(dataset_preparado_pd))
print("Registros recuperados:", len(df_verificacion))

dataset_preparado.unpersist(blocking=True)
spark.stop()
print("SparkSession detenida.")
```

```
Ruta de salida: C:\Users\julia\OneDrive\Escritorio\MCD\Tesis\proyecto_investigacion
\data\processed\apple_support_filtrado.parquet
Registros enviados: 232973
Registros recuperados: 232973
SparkSession detenida.
```